Generic FPSO Process Safety Case

Oil production with gas reinjection · Learning case for FPSO Barrier Lab · Frank Deezua Barilugbene · 12 October 2026

Summary

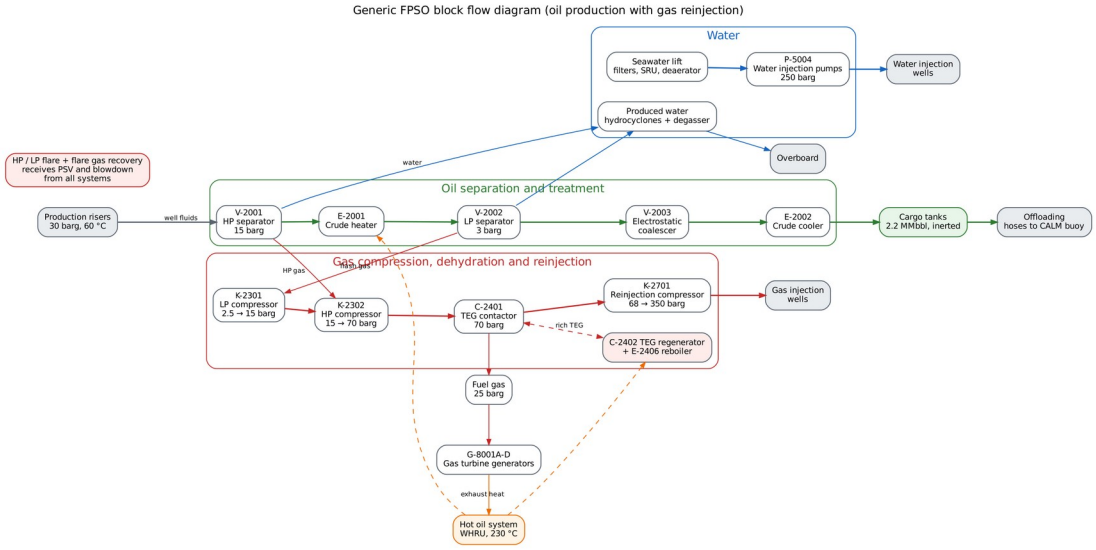
- A typical deepwater FPSO: oil goes to the cargo tanks, gas is dried and reinjected, and seawater is injected.
- Capacities match a public Erha-class FPSO. All operating conditions are typical values for learning.
- Your gas sweetening plant maps onto the FPSO's TEG dehydration unit, item for item.
- Four LOPA scenarios cover the classic FPSO hazards. In two of them, a simple LOPA that multiplies every layer passes, but the independence checks show it fails.
- Every LOPA number is calculated by the tool's engine and checked by `scripts/verify-fpso.mjs`.

Educational case. Not a representation of any operator's facility.

1. Basis of design

Item	Value
Oil processing	210,000 bbl/d
Gas reinjection	340 MMscf/d
Water injection	150,000 bbl/d
Crude storage	2.2 million bbl
Hull	285 m × 63 m, spread-moored
Water depth	1,200 m
People on board	100
Gas-oil ratio	About 1,620 scf/bbl (derived)
H2S / CO2	50 ppmv / 2 mol% (ASSUMED, slightly sour)
Riser arrival	30 barg, 60 °C (ASSUMED)

2. Process overview



Generic FPSO block flow diagram

- **Oil:** HP separator, crude heater, LP separator, electrostatic coalescer, crude cooler, cargo tanks, offloading.
- **Gas:** LP and HP compression, TEG dehydration, then reinjection at 350 barg. Fuel gas is taken off after drying.
- **Water:** produced water is cleaned and discharged overboard. Treated seawater is injected at 250 barg.
- **Heat:** gas turbine exhaust heats hot oil, which serves the crude heater and the TEG reboiler.
- **Flare:** HP and LP flares receive relief and blowdown from every system. Flare gas recovery keeps routine flaring at zero.

3. Key equipment and operating conditions

System	Tag	Equipment	Operating conditions (ASSUMED)
Oil separation and treatment	V-2001	HP separator (3-phase)	15 barg, 75 °C, design 25 barg
Oil separation and treatment	E-2001	Crude heater (hot oil)	75 → 85 °C
Oil separation and treatment	V-2002	LP separator	3 barg, 85 °C, design 10 barg
Oil separation and treatment	V-2003	Electrostatic coalescer	2.5 barg, 85 °C, BS&W < 0.5%
Oil separation and treatment	E-2002	Crude cooler (seawater)	85 → 45 °C
Gas compression (LP and HP)	V-2301	LP compressor suction scrubber	2.5 barg
Gas compression (LP and HP)	K-2301	LP compressor	2.5 → 15 barg
Gas compression (LP and HP)	V-2302	HP compressor suction scrubber	15 barg
Gas compression (LP and HP)	K-2302	HP compressor	15 → 70 barg
Gas compression (LP and HP)	E-2302	HP compressor aftercooler	to 35 °C
Gas dehydration (TEG)	V-2401	Contactor inlet scrubber	70 barg, 35 °C
Gas dehydration (TEG)	V-2402	Contactor inlet filter coalescer	70 barg

System	Tag	Equipment	Operating conditions (ASSUMED)
Gas dehydration (TEG)	C-2401	TEG contactor	70 barg, 35 °C
Gas dehydration (TEG)	V-2404	Rich TEG flash drum	4 barg
Gas dehydration (TEG)	E-2404	Glycol/glycol exchanger	Rich TEG heated to 150 °C
Gas dehydration (TEG)	C-2402	TEG regenerator (still column)	0.1 barg, design 3.5 barg
Gas dehydration (TEG)	E-2406	TEG reboiler (hot oil)	200 °C max
Gas dehydration (TEG)	E-2402	Still column reflux condenser	Overhead to 100 °C
Gas dehydration (TEG)	V-2406	Still vent KO drum / BTEX condenser	Near atmospheric
Gas dehydration (TEG)	P-2405A/B	Lean TEG circulation pumps	to 70 barg
Gas dehydration (TEG)	V-2405	TEG charcoal filter	-
Gas dehydration (TEG)	T-2401	TEG storage tank	-
Gas reinjection compression	V-2701	Reinjection compressor suction scrubber	68 barg
Gas reinjection compression	K-2701	Gas reinjection compressor	68 → 350 barg, design 385 barg discharge
Gas reinjection compression	E-2701	Reinjection aftercooler	to 50 °C
Gas reinjection compression	SDV-2701	Injection riser ESD valve	350 barg
Fuel gas	V-2801	Fuel gas KO drum	25 barg
Produced water treatment	A-4401	Hydrocyclones	13 barg inlet
Produced water treatment	V-4402	Degasser / induced gas flotation	0.5 barg
Seawater injection	P-5001	Seawater lift pumps	Lift to 6 barg
Seawater injection	A-5002	Sulphate removal unit (nanofiltration)	25 barg
Seawater injection	V-5003	Vacuum deaerator	Vacuum
Seawater injection	P-5004	Water injection pumps	to 250 barg
Hot oil system	H-4101	Waste heat recovery units	Hot oil to 230 °C
Hot oil system	V-4101	Hot oil expansion drum	N2 blanketed
Hot oil system	P-4101	Hot oil circulation pumps	-
Power generation	G-8001A-D	Gas turbine generators (4 × 25 MW)	Fuel gas or diesel
Flare	V-4301	HP flare KO drum	-
Flare	V-4302	LP flare KO drum	-
Flare	X-4301	Flare tower with HP and LP tips	-
Cargo tanks and offloading	T-6001-12	Cargo tanks	Atmospheric, inert gas blanket, 45 °C
Cargo tanks and offloading	P-6001	Cargo offloading pumps	-
Cargo tanks and offloading	X-6002	Offloading hose reel (starboard quarter)	-
Riser balcony and manifolds	SDV-1001	Production riser ESD valves	-

4. How your gas sweetening case maps onto the FPSO

TEG dehydration uses the same kit as amine sweetening: a high-pressure contactor, then a low-pressure regeneration loop. Your What-if, LOPA and bowtie carry over almost unchanged.

Your coursework item	FPSO equivalent	What changes
V-4401 Feed gas KO drum	V-2401 Contactor inlet scrubber	Same role
V-4402 Feed gas filter coalescer	V-2402 Contactor inlet filter coalescer	Same role
C-4401 Amine absorber	C-2401 TEG contactor	Removes water, not H2S/CO2
V-4404 Rich amine flash drum	V-2404 Rich TEG flash drum	Same role
E-4404A/B/C Rich/lean exchanger	E-2404 Glycol/glycol exchanger	Same role
C-4402 Amine regenerator	C-2402 TEG regenerator (still column)	Runs near atmospheric. TEG degrades above about 204 °C
E-4406A/B Reboiler	E-2406 TEG reboiler (hot oil)	Heated by hot oil from turbine exhaust
EA-4402 Reflux condenser	E-2402 Still column reflux condenser	Same role
V-4406 Reflux drum	V-2406 Still vent KO drum / BTEX condenser	Vent gas carries BTEX. Benzene exposure risk
P-4405A/B Reflux pumps	P-2405A/B Lean TEG circulation pumps	Pump lean TEG up to 70 barg
V-4405 Charcoal filter	V-2405 TEG charcoal filter	Same role
T-4401 Amine storage tank	T-2401 TEG storage tank	Same role
Inlet gas hot oil heater E-4402	Hot oil system (H-4101, E-2001)	Your hot oil HAZOP applies directly

5. Deck layout (stern to bow)

Zone (m from stern)	System
0 to 35	Accommodation, temporary refuge, helideck, lifeboats
35 to 60	Power generation
40 to 260	Cargo tanks and offloading
60 to 80	Hot oil system
80 to 105	Seawater injection
105 to 125	Produced water treatment
120 to 200	Riser balcony and manifolds
125 to 160	Oil separation and treatment
160 to 190	Gas compression (LP and HP)
185 to 195	Fuel gas
190 to 210	Gas dehydration (TEG)
210 to 235	Gas reinjection compression
255 to 275	Flare

The accommodation sits at the stern, the flare at the bow. Process modules sit between them, so the hazard rises as you move away from where people live. Risers hang off the port side balcony, and the cargo tanks run below deck for most of the hull.

6. Major hazards by system

System	Major hazards
Oil separation and treatment	Gas blowby from HP to LP separator. Loss of containment of hot crude. Separator overpressure on blocked outlet
Gas compression (LP and HP)	Liquid carryover into compressor. Compressor seal leak. Surge damage
Gas dehydration (TEG)	Regenerator overpressure (loss of cooling or blocked vent). Glycol/glycol exchanger tube rupture. Hydrocarbon carryover flashing in reboiler. Gas blowby from contactor to flash drum
Gas reinjection compression	Discharge overpressure on blocked outlet. High-pressure gas release and jet fire. Riser failure
Fuel gas	Overpressure of fuel gas system on let-down valve failure
Produced water treatment	Oil overboard (environmental). Gas blowby to degasser
Seawater injection	High-pressure water release (personnel injury). Chemical handling (biocide, oxygen scavenger)
Hot oil system	Hot oil leak above autoignition temperature. Process gas leaking into hot oil via tube rupture
Power generation	Fuel gas leak in turbine enclosure. Loss of main power
Flare	Flare KO drum liquid overfill (burning liquid rain-out). Thermal radiation during blowdown
Cargo tanks and offloading	Cargo tank overfill. Loss of inerting (explosive atmosphere). Hose rupture during offloading (spill)
Riser balcony and manifolds	Riser rupture with full inventory release. Dropped objects onto risers
Accommodation, temporary refuge, helideck, lifeboats	Smoke or gas ingress into temporary refuge. Helicopter accident

7. Risk criteria (example)

Severity	Description	Tolerable frequency (per year)
Catastrophic	Multiple fatalities or major escalation	1×10^{-6}
Major	Single fatality or major environmental release	1×10^{-5}
Serious	Lost time injury or significant release	1×10^{-4}

These are example targets for learning. Each company sets its own.

8. LOPA scenarios

Each scenario is calculated in two ways:

- **Naive:** multiplies every layer someone claimed.
- **Strict:** applies three CCPS-style checks:
  - a layer cannot share a sensor with the initiating event
  - a human layer needs at least 10 minutes to respond
  - a relief valve must be sized for the scenario

Layers that share a sensor with each other count as one layer.

Scenario	Severity	Case	Naive result (per year)	Strict result (per year)	Strict verdict
F1 HP separator gas blowby to LP separator	Catastrophic	As found	1.5×10^{-6}	1.5×10^{-3}	Fail (1,500× over)
F1 HP separator gas blowby to LP separator	Catastrophic	Fix	5×10^{-7}	5×10^{-7}	Pass (2× margin)
F2 Reinjection compressor discharge blocked	Catastrophic	As found	5×10^{-7}	5×10^{-6}	Fail (5× over)
F2 Reinjection compressor discharge blocked	Catastrophic	Fix	5×10^{-7}	5×10^{-7}	Pass (2× margin)
F3 TEG regenerator overpressure (from coursework LOPA-01)	Major	As found	1.5×10^{-6}	1.5×10^{-5}	Fail (1.5× over)
F3 TEG regenerator overpressure (from coursework LOPA-01)	Major	Fix	1.5×10^{-6}	1.5×10^{-6}	Pass (6.7× margin)
F4 Cargo tank overfill (Buncefield analogue)	Major	As found	1.5×10^{-5}	1.5×10^{-3}	Fail (150× over)
F4 Cargo tank overfill (Buncefield analogue)	Major	Fix	5×10^{-7}	5×10^{-7}	Pass (20× margin)

F1. HP separator gas blowby to LP separator

- **Cause:** HP separator level control loop fails (LT-1 or LCV fails open). Liquid level is lost and gas flows into the LP separator.
- **Consequence:** LP separator (design 10 barg) overpressured by HP gas. Rupture, fire and explosion in the separation module.
- **Severity:** Catastrophic. Target 1×10^{-6} per year.

As found. Initiating event: BPCS level loop failure, 0.1 per year (sensor LT-1).

Layer	PFD	Sensor	Strict check
Low level alarm LAL + operator closes inlet	0.1	LT-1	Rejected: Shares LT-1 with the initiating event
LSLL SIF closes liquid outlet SDV (SIL 1)	0.015	LT-2	Valid
LP separator PSV	0.01	-	Rejected: Not sized for this scenario

Naive: 1.5×10^{-6} per year. Strict: 1.5×10^{-3} per year, fail (1,500× over).

Fix. Initiating event: BPCS level loop failure, 0.1 per year (sensor LT-1).

Layer	PFD	Sensor	Strict check
Low level alarm from independent LT-3 + operator	0.1	LT-3	Valid
LSLL SIF upgraded to SIL 2	5×10^{-3}	LT-2	Valid
LP separator PSV resized for full gas blowby	0.01	-	Valid

Naive: 5×10^{-7} per year. Strict: 5×10^{-7} per year, pass (2× margin).

What this teaches:

- The low level alarm comes from the same transmitter as the control loop that failed. It cannot protect against its own failure.
- A PSV sized only for fire cannot be credited for gas blowby. Blowby needs a much larger relief rate.
- The fix needs an independent transmitter, a SIL 2 trip and a resized PSV.

F2. Reinjection compressor discharge blocked

- **Cause:** Injection riser ESD valve SDV-2701 closes spuriously while the compressor runs.
- **Consequence:** Discharge piping overpressured above 385 barg design. High-pressure gas release, jet fire and escalation.
- **Severity:** Catastrophic. Target 1×10^{-6} per year.

As found. Initiating event: Spurious closure of SDV-2701, 0.1 per year.

Layer	PFD	Sensor	Strict check
High pressure alarm PAH + operator trips compressor	0.1	PT-1	Rejected: Only 1 min to respond
PSHH SIF trips compressor (SIL 2)	5×10^{-3}	PT-2	Valid
Discharge PSV	0.01	-	Valid

Naive: 5×10^{-7} per year. Strict: 5×10^{-6} per year, fail (5× over).

Fix. Initiating event: Spurious closure of SDV-2701, 0.1 per year.

Layer	PFD	Sensor	Strict check
PAH kept as an operator aid, not credited	0.1	PT-1	Not claimed
SIL 3 high-integrity trip, 2oo3 transmitters	5×10^{-4}	PT-2/3/4	Valid
Discharge PSV	0.01	-	Valid

Naive: 5×10^{-7} per year. Strict: 5×10^{-7} per year, pass (2× margin).

What this teaches:

- Discharge pressure reaches the relief set point in about a minute. An operator cannot respond in time, so the alarm earns no credit.
- Naive LOPA passes. Strict LOPA fails by 5×.
- The fix is a SIL 3 high-integrity trip with 2oo3 voting, plus the PSV. Keep the alarm as an aid.

F3. TEG regenerator overpressure (from coursework LOPA-01)

- **Cause:** Loss of reflux condenser cooling or blocked still vent while the reboiler fires.
- **Consequence:** Regenerator C-2402 overpressured. Release of hot TEG and hydrocarbon vapour, fire in the dehydration module.
- **Severity:** Major. Target 1×10^{-5} per year.

As found. Initiating event: Loss of overhead cooling or blocked vent, 0.1 per year.

Layer	PFD	Sensor	Strict check
High pressure alarm PAH + operator	0.1	PT-1	Shares a sensor with SIF. Counted once
PSHH cuts hot oil to reboiler (SIL 1)	0.015	PT-1	Shares a sensor with ALM. Counted once
Regenerator PSV	0.01	-	Valid

Naive: 1.5×10^{-6} per year. Strict: 1.5×10^{-5} per year, fail (1.5× over).

Fix. Initiating event: Loss of overhead cooling or blocked vent, 0.1 per year.

Layer	PFD	Sensor	Strict check
High pressure alarm PAH + operator	0.1	PT-1	Valid
PSHH on dedicated PT-2 (SIL 1)	0.015	PT-2	Valid
Regenerator PSV	0.01	-	Valid

Naive: 1.5×10^{-6} per year. Strict: 1.5×10^{-6} per year, pass (6.7× margin).

What this teaches:

- Your coursework scenario on the FPSO. It is the same mechanism, but the consequence is a fire, not toxic H2S, so the target is 1×10^{-5}.
- The alarm and trip share PT-1. Naive LOPA passes, but strict LOPA fails by 1.5×.
- The fix is cheap: a dedicated transmitter for the trip.

F4. Cargo tank overfill (Buncefield analogue)

- **Cause:** Radar level gauge RLG-1 sticks, or wrong tank lineup during changeover.
- **Consequence:** Crude overflows through P/V valves onto the main deck. Pool fire, vapour cloud, oil to sea.
- **Severity:** Major. Target 1×10^{-5} per year.

As found. Initiating event: Gauge failure or lineup error, 0.1 per year (sensor RLG-1).

Layer	PFD	Sensor	Strict check
High level alarm from the radar gauge	0.1	RLG-1	Rejected: Shares RLG-1 with the initiating event
High-high level alarm from switch LS-2	0.1	LS-2	Shares a sensor with SIF. Counted once
Overfill trip closes tank inlet (SIL 1)	0.015	LS-2	Shares a sensor with HHA. Counted once

Naive: 1.5×10^{-5} per year. Strict: 1.5×10^{-3} per year, fail (150× over).

Fix. Initiating event: Gauge failure or lineup error, 0.1 per year (sensor RLG-1).

Layer	PFD	Sensor	Strict check
High-high alarm from independent switch LS-3	0.1	LS-3	Valid
Overfill trip on LS-2 upgraded to SIL 2	5×10^{-3}	LS-2	Valid
Overflow line to slop tank (passive)	0.01	-	Valid

Naive: 5×10^{-7} per year. Strict: 5×10^{-7} per year, pass (20× margin).

What this teaches:

- The high level alarm uses the same radar gauge that failed. At Buncefield the gauge stuck and the independent switch was inoperable.
- The high-high alarm and the trip share switch LS-2, so only one counts.
- The fix needs an independent switch, a SIL 2 trip and a passive overflow line to the slop tank.

9. Personnel safety tasks

These tasks cover the HSE advisory side of the role.

Task	Permit	Key controls
Hot work (welding) on a pipe support in the separation module	Hot work permit	Gas test before and during work. Isolate and purge nearby lines. Fire watch. Check SIMOPS conflicts (no venting, no offloading start)
Confined space entry into a cargo tank for inspection	Confined space entry permit	Gas free and ventilate (O2, LEL, H2S, benzene). Isolate inert gas and cargo lines with blinds. Standby person and rescue plan. Continuous monitoring
Crane lift of a compressor spare over live plant	Lifting plan (critical lift)	Lift plan and weather limits. Exclusion zone. Avoid lifting over risers and live hydrocarbon lines. Competent banksman
Tandem offloading to a shuttle tanker via CALM buoy	Offloading procedure	Weather window. Hose inspection and pressure test. Emergency release coupling check. No hot work or helicopter ops during offloading (SIMOPS)

10. Assumptions to check

- Operating pressures and temperatures are typical values. They are not from any real facility.
- H2S at 50 ppmv is assumed so the H2S lessons still apply.
- The initiating event frequencies for F2 and F4 are ASSUMED generic values.
- The example risk criteria are for learning. Know why you'd use them.

11. Next steps

1. Review this case and change anything you disagree with.
2. Build the 3D FPSO. Each system is a clickable module that opens its hazards and LOPA.
3. Add the independence toggle. Re-assign a sensor and watch the verdict flip.
4. Add case studies: Buncefield on the cargo tanks, Piper Alpha on permits, Cidade de São Mateus on a pump room leak.